

Notes: De Loecker (Econometrica, 2011)

Product Differentiation, Multiproduct Firms, and Estimating the Impact of Trade Liberalization on Productivity

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1 Big picture

- **Question.** What is the true productivity effect of trade liberalization once we **separate physical efficiency** from price and demand responses?
- **Setting.** Belgian textile firms selling mainly into the EU-15 market, 1994–2002, during a period of major quota liberalization under the Multi-Fibre Arrangement and the Europe Agreements.
- **Core problem.** Most production-function studies use deflated revenue instead of physical output. If firm-level prices differ from the industry price index and are correlated with input choices, estimated production coefficients and “productivity” measures are biased.
- **Main idea.** Embed a demand system inside the production-function framework. Use product mix, segment demand, product/product-group controls, and firm-specific quota exposure to distinguish productivity from prices and demand.
- **Main result.** Standard approaches imply that removing all quotas raises firm productivity by about 8% (and as much as 13.5% in long differences). After correcting for unobserved prices and demand, the average effect falls to about 2%, or about 4.5% in long differences.
- **Broader lesson.** Trade reform affects not only efficiency, but also residual demand, prices, and markups. Revenue productivity is therefore not a clean measure of physical productivity in differentiated-product industries.

2 Data and measurement (key objects)

2.1 Production data

- **Source.** BELFIRST / National Bank of Belgium administrative balance-sheet data.
- **Coverage.** Belgian textile producers, 1994–2002.
- **Variables.** Revenue, value added, employment, capital, investment, and intermediate inputs, plus entry and exit.
- **Summary facts (Table III).** Average employment is around 90 workers per firm. Average sales rise from about 3.2 to 4.5 million euros. Value added per worker rises from 45.1 to 53.7, while the textile producer price index is roughly flat to declining.
- **Interpretation.** Raw “productivity” and declining prices move together, already suggesting that standard revenue-based productivity measures mix quantity and price effects.

2.2 Product data and market structure

- **Source.** FEBELTEX product listings for roughly 310 Belgian textile firms.
- **Matched sample.** 308 firms matched to production data, covering about 75% of total textile employment.
- **Product space.** 563 products grouped into five segments: Interior, Clothing, Technical Textiles, Finishing, and Spinning/Preparing.

- **Multiproduct prevalence.** Firms produce on average 9 products; half produce 3 or fewer. About 25% are active in more than one segment, and these firms are about 20% larger.
- **Tables IV and V.** Most firms are concentrated in one segment, but Technical and Finishing firms are more diversified. Product-level specialization is even stronger than firm-level segment participation suggests.

2.3 Trade protection data

- **Source.** European Commission’s SIGL database on product-country-year textile quotas.
- **Coverage.** 182 quota product categories and 56 supplying countries.
- **Trade policy event.** The number of quota protections falls from 1,046 in 1994 to 486 in 2002, a decline of 54% (Table I). At the same time, quota levels on remaining restricted products rise, indicating both abolition and loosening.
- **Import reallocation.** Table II shows a large shift toward China, Turkey, Romania, Bangladesh, and other lower-wage suppliers.
- **Why this matters.** Belgian firms face changing residual demand and price pressure as EU protection falls and import composition shifts.

2.4 Key protection variable

- **Product-country quota dummy.** $q_{ect}^r = 1$ if product c from exporter e is quota restricted in year t .
- **Product-level quota rate.** qr_{ct} is the weighted average across exporters, with weights based on supplier production potential, not quota size.
- **Firm-level quota exposure.** q_{it}^r averages qr_{ct} across the firm’s products. It lies in $[0, 1]$: 0 means no product protected; 1 means every product protected against all supplying countries.
- **Figure 1.** Average quota restriction declines in all five segments, with meaningful variation across segments. That cross-segment variation is crucial for identifying segment-specific demand elasticities.

3 Models and empirical specifications (all equations + interpretation)

3.1 Single-product benchmark

The physical production function is Cobb–Douglas:

$$Q_{it} = L_{it}^{\alpha_l} M_{it}^{\alpha_m} K_{it}^{\alpha_k} \exp(\omega_{it} + u_{it}). \quad (1)$$

- **Interpretation.** ω_{it} is true firm productivity; u_{it} is measurement error / transitory noise.

- **Problem.** Physical output Q_{it} is not observed, so standard studies replace it with deflated sales and thereby import firm-level price variation into productivity measurement.

Demand for a firm's product in segment s is

$$Q_{it} = Q_{st} \left(\frac{P_{it}}{P_{st}} \right)^{\eta_s} \exp(\xi_{it}). \quad (2)$$

- **Interpretation.** Q_{st} is a segment demand shifter, P_{st} a segment price index, and ξ_{it} an unobserved demand shock.
- **Markup link.** Under CES demand and monopolistic competition, the segment Lerner index is $\beta_s = 1/|\eta_s|$.

Using $R_{it} = P_{it}Q_{it}$, revenue can be written as

$$R_{it} = Q_{it}^{(\eta_s+1)/\eta_s} Q_{st}^{-1/\eta_s} P_{st} \exp(-\xi_{it}/\eta_s). \quad (3)$$

Defining deflated revenue $\tilde{r}_{it} = r_{it} - p_{st}$, the estimating equation becomes

$$\tilde{r}_{it} = \beta_l l_{it} + \beta_m m_{it} + \beta_k k_{it} + \beta_s q_{st} + \omega_{it}^* + \xi_{it}^* + u_{it}. \quad (4)$$

- **Reduced-form coefficients.** $\beta_h = \frac{\eta_s+1}{\eta_s} \alpha_h$ for $h \in \{l, m, k\}$.
- **Key identification object.** β_s identifies inverse demand elasticity / markup.
- **Main lesson.** Standard production-function coefficients are demand-contaminated reduced-form parameters, not pure technology parameters.

3.2 Multiproduct extension

For product j of firm i , production is

$$Q_{ijt} = (c_{ijt}L_{it})^{\alpha_l} (c_{ijt}M_{it})^{\alpha_m} (c_{ijt}K_{it})^{\alpha_k} \exp(\omega_{it} + u_{it}), \quad (5)$$

and with proportional input allocation across J_{it} products,

$$Q_{ijt} = J_{it}^{-\gamma} Q_{it}, \quad \gamma = \alpha_l + \alpha_m + \alpha_k. \quad (6)$$

This yields the firm-level sales equation

$$\tilde{r}_{it} = \beta_{np} np_{it} + \beta_l l_{it} + \beta_m m_{it} + \beta_k k_{it} + \beta_s q_{st} + \omega_{it}^* + \xi_{it}^* + u_{it}, \quad (7)$$

where $np_{it} = \ln(J_{it})$. With multiple segments,

$$\tilde{r}_{it} = \beta_{np} np_{it} + \beta_l l_{it} + \beta_m m_{it} + \beta_k k_{it} + \sum_{s=1}^S s_{is} \beta_s q_{st} + \omega_{it}^* + \xi_{it}^* + u_{it}. \quad (8)$$

- **Why this matters.** Product mix lets the author estimate different demand elasticities by segment instead of imposing one industry-wide markup.
- **Trade-off.** This requires proportional input allocation across products and the assumption that the product mix is observed well enough to recover segment exposure.

3.3 Quota exposure

The average quota restriction for product c is

$$qr_{ct} = \sum_e a_{et} q_{ect}^r, \quad (9)$$

and the firm-level protection measure is

$$q_{it}^r = \sum_{c \in J(i)} a_c qr_{ct}. \quad (10)$$

- **Interpretation.** q_{it}^r is the firm's product-weighted exposure to quota protection.
- **Economic role.** Quotas affect firm residual demand and prices contemporaneously, and productivity with a lag.

3.4 Main estimating equation with demand controls

Unobserved demand is decomposed into product, product-group, and protection components, leading to the main specification:

$$\tilde{r}_{it} = \beta_l l_{it} + \beta_m m_{it} + \beta_k k_{it} + \beta_s q_{st} + \sum_{j \in J(i)} \delta_j D_{ij} + \sum_{g \in G(i)} \delta_g D_{ig} + \tau q_{it}^r + \omega_{it}^* + \varepsilon_{it}. \quad (11)$$

- **Interpretation.** Product and product-group fixed effects absorb persistent demand differences across products; q_{it}^r captures firm-specific protection-driven demand shifters.
- **Why this is crucial.** The omitted-price-variable bias is not handled by standard OP/LP procedures unless these demand controls are brought in explicitly.

3.5 Productivity law of motion and timing

The paper imposes

$$\omega_{it} = g_t(\omega_{i,t-1}, q_{i,t-1}^r) + v_{it}. \quad (12)$$

- **Timing assumption.** Current quotas shift residual demand and prices immediately, but productivity only changes with a lag.
- **Interpretation.** Firms need time to reorganize production, cut slack, adopt better management, or exploit product-mix changes after liberalization.

3.6 Levinsohn–Petrin style material-control approach

Material demand is

$$m_{it} = m_t(k_{it}, \omega_{it}, q_{it}^r, q_{st}, D), \quad (13)$$

which can be inverted as

$$\omega_{it} = h_t(k_{it}, m_{it}, q_{it}^r, q_{st}, D). \quad (14)$$

The first stage is

$$\tilde{r}_{it} = \beta_l l_{it} + \phi_t(m_{it}, k_{it}, q_{it}^r, q_{st}, D) + \varepsilon_{it}, \quad (15)$$

with $\phi_t(\cdot) = \beta_m m_{it} + \beta_k k_{it} + \beta_s q_{st} + \delta D + \tau q_{it}^r + h_t(\cdot)$. The second-stage moments are

$$\mathbb{E} \left[v_{i,t+1}(\beta_m, \beta_k, \beta_s, \tau, \delta) \begin{pmatrix} m_{it} \\ k_{i,t+1} \\ q_{st} \\ q_{i,t+1}^r \\ D \end{pmatrix} \right] = 0. \quad (16)$$

- **Interpretation.** Static input demand controls for unobserved productivity without requiring a full dynamic investment problem.
- **Key benefit.** This is especially attractive here because quotas enter as an additional state variable in the firm problem.

3.7 Olley–Pakes style investment-control approach

Investment satisfies

$$i_{it} = i_t(k_{it}, \omega_{it}, q_{it}^r, D), \quad (17)$$

which is inverted to

$$\omega_{it} = h_t(k_{it}, i_{it}, q_{it}^r, D). \quad (18)$$

The first stage becomes

$$\tilde{r}_{it} = \beta_l l_{it} + \beta_m m_{it} + \beta_s q_{st} + \phi_t(i_{it}, k_{it}, q_{it}^r, D) + \varepsilon_{it}, \quad (19)$$

and the second stage uses

$$\mathbb{E} \left[v_{i,t+1}(\beta_k, \tau, \delta) \begin{pmatrix} k_{i,t+1} \\ q_{i,t+1}^r \\ D \end{pmatrix} \right] = 0. \quad (20)$$

- **Interpretation.** This is the dynamic-control analogue of the material-proxy approach.
- **Caveat.** The quota process now enters the state space and therefore requires extra assumptions for invertibility.

3.8 Productivity measures and the bias in standard approaches

Corrected productivity is recovered as

$$\hat{\omega}_{it} = \left(\tilde{r}_{it} - \hat{\beta}_l l_{it} - \hat{\beta}_m m_{it} - \hat{\beta}_k k_{it} - \hat{\beta}_s q_{st} - \hat{\tau} q_{it}^r \right) \frac{\hat{\eta}_s}{\hat{\eta}_s + 1}. \quad (21)$$

Standard revenue-based productivity is

$$\hat{\omega}_{it}^{st} = \tilde{r}_{it} - x_{it} \hat{\beta}, \quad (22)$$

whereas true physical productivity is conceptually

$$\hat{\omega}_{it} = q_{it} - x_{it} \hat{\alpha}. \quad (23)$$

Their difference is

$$\hat{\omega}_{it}^{st} - \hat{\omega}_{it} = (p_{it} - p_t) + x_{it}(\hat{\alpha}/|\hat{\eta}|). \quad (24)$$

- **Interpretation.** Standard productivity mixes firm-specific prices with the technology term and also uses the wrong scale correction.
- **Key empirical implication.** If liberalization lowers prices and changes demand, standard measures mechanically attribute too much of that to productivity.

3.9 Why standard trade-productivity regressions are biased

The common two-stage approach regresses standard productivity on quotas:

$$\omega_{it}^{st} = c + \lambda q_{it}^r + e_{it}, \quad (25)$$

while a single-equation version writes

$$\tilde{r}_{it} = \beta_l l_{it} + \beta_m m_{it} + \beta_k k_{it} + \lambda q_{it}^r + e_{it}^*. \quad (26)$$

- **Why both fail.** They require that quotas affect productivity but do not affect prices, residual demand, or returns to scale — exactly the restriction the paper rejects.

3.10 Alternative demand system

As a robustness check, the paper shows the framework is consistent with a logit-style discrete-choice demand system:

$$V_{ljt} = \delta_j D_j + \eta p_{jt} + \xi_{jt} + \zeta_{ljt}, \quad (27)$$

which implies

$$\ln(ms_{jt}) - \ln(ms_{0t}) = \delta_j D_j + \eta p_{jt} + \xi_{jt}, \quad (28)$$

and therefore yields the same revenue-production estimating form,

$$r_{it} = \beta_0 + \beta_l l_{it} + \beta_m m_{it} + \beta_k k_{it} + \beta_s q_{st} + \delta_i D_i + \omega_{it}^* + \xi_{it}^* + u_{it}. \quad (29)$$

- **Interpretation.** The main message is not tied to one exact demand system. What matters is separating price and demand from productivity.

4 Identification and instruments (what makes the estimates credible)

4.1 Core identifying variation

- **Exogenous quota changes.** Textile quotas are decided at the EU level, so any one Belgian producer plausibly has negligible influence on trade policy.
- **Current demand vs. lagged productivity.** Current quotas affect residual demand and prices immediately; productivity responds with a lag through $g_t(\omega_{i,t-1}, q_{i,t-1}^r)$. This timing distinction is the paper's main identification device.
- **Observed demand shifters.** Segment output q_{st} , product and product-group indicators, and firm-specific quota exposure q_{it}^r absorb much of the price and demand variation that would otherwise contaminate productivity.

4.2 Why proxy estimators help but are not enough on their own

- **Simultaneity.** OP/LP address endogenous input choice by conditioning on an investment or material proxy for productivity.
- **What they miss.** Standard proxy estimators do *not* solve the omitted-price problem by themselves, because firms in differentiated-product industries face firm-specific prices that co-move with inputs.
- **Contribution of this paper.** Augment proxy estimation with demand-side structure so that one can recover both markups and physical productivity.

4.3 Important assumptions / caveats

- **Time-invariant product mix.** Product data are observed in a cross section, so the analysis assumes firms' product mix does not adjust much over time. The paper argues this is plausible in its 9-year panel and shows little mix adjustment in a knitwear subsample.

- **Interpretation if product mix changes.** Then estimated productivity partly captures product-mix responses to trade reform. The paper explicitly notes that it cannot cleanly separate pure efficiency changes from product reallocation.
- **Persistent demand shocks.** If demand shocks share the same Markov structure as productivity, they can be absorbed into the control function, but then “productivity” would again mix demand with technology.

5 Tables and figures

5.1 Table I: Number of quotas and quota levels

Read:

- The number of product-country quota protections drops from 1,046 in 1994 to 486 in 2002, a decline of 54%.
- The number of quota categories measured in kilograms falls 44%, and those measured in pieces fall 60%.
- At the same time, the average quota level on remaining restricted products rises substantially, by 72% for kg quotas and 44% for piece quotas.
- The combination means liberalization happened both through abolition and through larger remaining quotas.

TABLE I
NUMBER OF QUOTAS AND AVERAGE QUOTA LEVELS (IN MILLIONS)

	Number of Quota Protections	kg		No. of Pieces	
		No. of Quotas	Level	No. of Quotas	Level
1994	1,046	466	3.10	580	8.58
1995	936	452	3.74	484	9.50
1996	824	411	3.70	413	7.95
1997	857	413	3.73	444	9.28
1998	636	329	4.21	307	9.01
1999	642	338	4.25	304	10.53
2000	636	333	4.60	303	9.77
2001	574	298	5.41	276	11.06
2002	486	259	5.33	227	12.37
Change	-54%	-44%	72%	-60%	44%

TABLE II
TOP SUPPLIERS OF TEXTILE PRODUCTS TO THE EU^a

Country	Top in 1995 (63%)		Country	Top in 2002 (68.3%)	
	Share	No. of Quotas		Share	No. of Quotas
China	10.6	66	China	15.9	35
Turkey	9.4	35	Turkey	12.5	0
India	6.5	19	India	5.6	17
Hong Kong	5.7	32	Romania	5.4	0
Poland	4.2	18	Tunisia	4.4	0
Tunisia	4.1	2	Bangladesh	4.0	3
United States	4.1	0	Morocco	3.8	0
Switzerland	4.0	0	Poland	3.4	0
Morocco	3.8	0	Hong Kong	3.3	18
Indonesia	3.3	14	Indonesia	2.8	12
Pakistan	2.7	15	Pakistan	2.7	14
Bangladesh	2.3	3	Czech Republic	2.3	0
Romania	2.3	30	Republic of Korea	2.2	27
Total imports		45,252			71,414
Imports top 13		28,509			48,776

^aTotal imports are in billions of euros and are from the author's own calculations and EUROSTAT.

5.2 Table II: Top suppliers of textile products to the EU

Read:

- China's import share rises from 10.6% to 15.9% while its quota count falls from 66 to 35.
- Turkey rises from 9.4% to 12.5% and faces zero quotas by 2002.
- Romania and the Czech Republic gain importance as EU candidate-country integration proceeds.

- This table shows that liberalization changed not just the level but also the composition of foreign competition facing Belgian producers.

5.3 Table III: Summary statistics of production data

Read:

- Average employment is stable to mildly rising, but value added per worker increases from 45.1 to 53.7.
- The producer price index is essentially flat, even slightly below the 1994 level for several years.
- Rising revenue-per-input measures alongside weak prices already suggests the need to distinguish physical productivity from pricing effects.

TABLE III
SUMMARY STATISTICS OF PRODUCTION DATA^a

Year	Employment	Total Sales	VA p/w	Materials	PPI
1994	89	3,185	45.1	13,160	100.00
1995	87	3,562	44.4	14,853	103.40
1996	83	3,418	45.2	14,313	99.48
1997	85	4,290	53.0	16,688	99.17
1998	90	4,482	50.9	17,266	98.86
1999	88	4,248	51.6	15,546	98.77
2000	90	4,763	52.5	17,511	102.98
2001	92	4,984	51.5	17,523	102.67
2002	99	4,500	53.7	17,053	102.89

^aEmployment is average number of full time employees, total sales is given in millions of euros, VA p/w is value added per worker, and materials is given in thousands of euros.

TABLE IV
NUMBER OF FIRMS ACROSS DIFFERENT SEGMENTS^a

	Interior	Clothing	Technical	Finishing	Spinning
Interior	77.0	4.8	15.8	7.3	1.8
Clothing		58.9	33.9	7.1	1.8
Technical			35.1	19.6	17.5
Finishing				39.6	12.5
Spinning					47.5
No. of firms	165	56	97	48	40

^aCells do not sum to 100 percent by row or column, as a firm can be active in more than two segments.

5.4 Tables IV and V: Segment participation and product mix

Read:

- Most firms are concentrated in a single segment, but multi-segment firms are not rare.
- Technical and Finishing firms appear less specialized, which helps identify segment demand elasticities using within-firm variation in exposure.
- The product-level data therefore do more than describe scope; they discipline the demand side of the model.

5.5 Figure 1: Evolution of quota restriction measure by segment

Read:

- All five segments show a clear decline in average quota protection over 1994–2002.
- The decline is not identical across segments, which is important because that variation identifies η_s .
- The figure also makes clear why a single aggregate demand parameter is too restrictive.

TABLE V
FIRMS' PRODUCT MIX ACROSS DIFFERENT SEGMENTS

	Interior	Clothing	Technical	Finishing	Spinning
Interior	83.72	2.78	8.27	4.41	0.80
Clothing	3.03	79.28	15.36	1.86	0.48
Technical	7.01	8.97	70.16	9.06	4.79
Finishing	5.75	3.52	15.53	72.85	2.35
Spinning	3.72	0.65	27.20	7.40	61.04
Median	2	6	8	11	9
Min	1	2	1	2	1

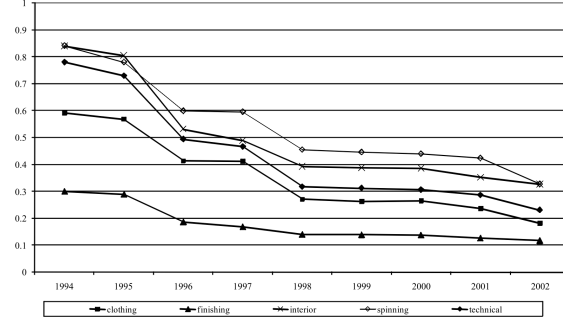


FIGURE 1.—Evolution of quota restriction measure by segment (1994–2002).

5.6 Table VI: Production function estimates

Read:

- OLS and Klette–Griliches (difference) estimates are benchmark revenue-based specifications.
- The augmented model shows that correcting simultaneity alone is not enough; unobserved price variation also matters.
- The industry-level demand coefficient in the augmented specification is about 0.309, implying an elasticity around -3.24 .
- The similar single-segment estimate (0.260, elasticity about -3.85) suggests the extra assumptions for multi-segment firms are not driving results.

TABLE VI
PRODUCTION FUNCTION ESTIMATES^a

Coefficient on	[1] OLS	[2] KG		[3] Proxy	[4] Augmented		[5]: Single Segment
	β	β	α	β	β	α	β
Labor	0.230 (0.009)	0.245 (0.0120)	0.334 (0.034)	0.211 (0.011)	0.213 (0.011)	0.308 (0.062)	0.230 (0.017)
Materials	0.630 (0.007)	0.596 (0.013)	0.812 (0.052)	0.628 (0.009)	0.627 (0.008)	0.906 (0.175)	0.650 (0.013)
Capital	0.088 (0.007)	0.019 (0.011)	0.026 (0.014)	0.093 (0.008)	0.104 (0.006)	0.150 (0.034)	0.090 (0.006)
Output		0.266 (0.046)			0.309 (0.134)		0.260 (0.137)
η		-3.76			-3.24		-3.85
No. of obs	1,291	1,291		985/1,291	985		735

^a Bootstrapped standard errors are given in parentheses. Results under [3] Proxy are obtained using the intermediate inputs as a proxy as suggested by Levinsohn and Petrin (2003) and are compared with the Olley and Pakes (1996) approach. The LP estimator was implemented using their estimator in STATA (lvpct) and the Wooldridge (2011) version of LP using ivreg2 in STATA.

TABLE VII
SEGMENT-SPECIFIC DEMAND PARAMETERS AND RETURNS TO SCALE^a

Demand Controls	Estimated Coefficient (β_s)		Implied Elasticity (η_s)	
	Not Included	Included	Not Included	Included
Industry (I)	0.35	0.26	-2.86	-3.85
Interior ($s = 1$)	0.24	0.16	-4.17	-6.25
Clothing ($s = 2$)	0.35	0.23	-2.86	-4.35
Technical ($s = 3$)	0.31	0.21	-3.23	-4.76
Finishing ($s = 4$)	0.33	0.22	-3.03	-4.55
Spinning ($s = 5$)	0.26	0.18	-3.85	-5.56
RTS	1.3	1.11		

^a Demand controls are product, product-group, and protection rate variables, (ξ_j, ξ_g, q_{it}). All coefficients are significant at the 1 percent level.

5.7 Table VII: Segment-specific demand parameters and returns to scale

Read:

- Without demand controls, implied elasticities range roughly from -2.86 to -4.17 .
- Once product, product-group, and protection controls are included, demand becomes substantially more elastic: Interior -6.25 , Clothing -4.35 , Technical -4.76 , Finishing -4.55 , Spinning -5.56 , with industry-wide -3.85 .
- Returns to scale fall from 1.30 to 1.11 once those demand controls are included.
- Interpretation: the omitted-price-variable bias was pushing down estimated elasticities and distorting the inferred technology parameters.

5.8 Table VIII: Impact of trade liberalization on productivity

Read:

- Standard OLS / proxy approaches imply large gains from quota removal: estimates around -0.161 , -0.153 , and -0.135 .
- Once the proxy estimator is adjusted so quotas affect productivity dynamics but prices are still ignored, the mean effect falls to about -0.086 .
- In the paper's corrected approach, the average quota effect is only about -0.021 , with support $[-0.27, 0.10]$.
- In long differences, the corrected effect is about -0.046 , compared with -0.135 using standard measures.
- Bottom line: standard measures overstate the productivity gains from liberalization by a factor of roughly 3 to 4.

TABLE VIII
IMPACT OF TRADE LIBERALIZATION ON PRODUCTIVITY^a

Approach	Description	Estimate	Support
I	OLS levels	-0.161^* (0.021)	n.a.
II.1	Standard proxy-levels	-0.153^* (0.021)	n.a.
II.2	Standard proxy-LD	-0.135^* (0.030)	n.a.
III	Adjusted proxy	-0.086 (0.006)	$[-0.129, -0.047]$
IV	Corrected	-0.021 sd: 0.067	$[-0.27, 0.100]$
V	Corrected LD	-0.046^{**} (0.027)	n.a.

^aI report standard errors in parentheses for the regressions, while I report the standard deviation (sd) of the estimated nonparametric productivity effect in my empirical model (given by $g(\cdot)$). * and ** denote significant at 5 or lower and 10 percent, respectively. LD refers to a 3 year differencing of a two-stage approach where Approach II.2 relies on standard productivity measures, as opposed to Approach V, which relies on my corrected estimates of productivity.

TABLE IX
ESTIMATED DEMAND PARAMETERS UNDER ALTERNATIVE SPECIFICATIONS^a

Specification	Segment					
	Industry	Interior	Clothing	Technical	Finishing	Spinning
Baseline	0.26	0.16	0.23	0.21	0.22	0.18
Flexible	0.22	0.14	0.23	0.19	0.22	0.21
Value added	0.29	0.20	0.21	0.27	0.25	0.21
Unrestricted	$\sigma = 0.008$ (0.113)	0.17	0.25	0.21	0.23	0.19

^aBaseline specification is compared to (i) Flexible, which allows for a general production function that is log additive in productivity, (ii) to a Value added production function specification, and (iii) to an unrestricted nested CES demand system. For the latter, I report the estimated cross-segment correlation coefficient σ and its standard error. I suppress the bootstrapped standard errors on the demand parameters, since they are all significant at the 1 percent level.

5.9 Table IX: Alternative specifications

Read:

- Allowing a more flexible production function or using value added leaves the segment demand parameters in a similar range.
- The unrestricted nested-CES estimate of the cross-segment correlation parameter is $\sigma = 0.008$ with a large standard error, so the data do not reject the restricted specification.
- The message of the paper therefore does not depend on one knife-edge functional-form assumption.

6 Short summary

- **Conclusion.** Trade liberalization in Belgian textiles did raise productivity, but much less than standard revenue-based methods suggest.
- **Intuition.** When quotas fall, firms face more competition, which changes prices and residual demand immediately. If we fail to net those out, we mistake demand/price responses for productivity gains.
- **Empirical lesson.** Estimating productivity in differentiated-product industries requires combining demand and production, not just running a production function on deflated sales.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Removing all quota protection raises firm-level productivity by only about 2% on average, instead of the 8% suggested by standard measures.
- **Magnitude.** In long differences, the corrected estimate is about 4.5% versus 13.5% using standard productivity.
- **Credibility.** The identification rests on exogenous quota reforms, explicit demand shifters, and proxy methods that handle simultaneity while separating price and productivity responses.
- **Mechanism evidence.** Quota reductions are associated with lower prices and more elastic demand, not just higher efficiency.
- **Quantitative implication.** Only a relatively small fraction of measured gains from liberalization should be interpreted as true physical productivity growth.
- **Additional discipline.** Results are robust to flexible production functions, value-added specifications, and a richer nested demand system.

7.2 Economic intuition

- Revenue productivity rises after liberalization for two conceptually distinct reasons: firms may truly become more efficient, or they may simply move along a new residual demand curve and charge different prices.
- In differentiated-product industries, those two forces are tightly intertwined because markups and prices vary systematically across firms and products.
- The paper’s key contribution is to separate them empirically, showing that a large share of the conventional “productivity” gain is actually a price/demand effect.
- The product-mix data are essential because firms are multiproduct and segments differ in demand elasticities, so a one-markup model is too coarse.

7.3 Takeaway

This paper is one of the foundational contributions showing that productivity estimation with revenue data is fundamentally a joint demand-and-supply problem. The empirical punchline is simple but powerful: standard trade-productivity regressions in differentiated-product industries can dramatically overstate efficiency gains because they confound physical productivity with prices, markups, and demand. For anyone working on trade reform, firm performance, or markups, the paper's main lesson is to treat revenue productivity with caution and to model the demand side explicitly whenever prices are not observed.